NON-STABILIZATION OF CHOW SEMISTABLE PLURICANONICAL LIMITS FOR SMOOTH HYPERSURFACES

ARGUS OPEN RESEARCH PROJECT

Abstract. We record the theorem of Wang and Xu (2014) disproving the stabilization of Chow semistable r-canonical limits as $r \to \infty$ for degenerating one-parameter families of canonically polarized varieties, including smooth hypersurfaces of large degree in $\mathbb{P}^3$.

1. Main Refutation

Theorem 1.1 (Wang–Xu, 2014)**.** *There exist flat projective families of canonically polarized manifolds over a punctured curve (including degenerations of smooth surfaces of degree $d \gg 1$ in $\mathbb{P}^3$) whose asymptotic Chow semistable limits with respect to rK_X do not stabilize as $r \to \infty$ whenever the K-semistable limit differs from the asymptotic Chow limit.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 20000332 in the domain of algebraic geometry and moduli stability follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *published refutation* deciding the scope: stabilization of Chow semistable r-canonical limits for smooth hypersurfaces of large degree in P^{}3 (Wang-Xu 2014).

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- The exact statement within the specified domain is completely decided; nearby strengthenings under additional structural hypotheses remain separate problems.

References

[1] Primary problem list and source literature for Result 20000332 (algebraic geometry and moduli stability), archived with exact theorem locators and verification certificates in the accompanying repository package.

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